24 Spr. 高代下 final (H). (回忆版)

Problem 1. Let X be the set of all symmetric $n\times n$ matrices over R, and $Y\subseteq X$ the subset of all positive semidefinite matrices.

(i) For two matrices $A,B\in X$, define $A\geq B$ if $A-B\in Y$. Prove that $\geq$ is a partial order, i.e. (1) $A\geq A$, (2) $A\geq B$ and $B\geq A$ implies $A=B$, (3) $A\geq B$ and $B\geq C$ implies $A\geq C$.

(ii) Prove that Y is convex, i.e. for $A,B\in Y$, we have $tA+(1-t)B\in Y$ for all real $t\in[0,1]$.

(iii) Let $A,B\in Y$. Suppose that $AB=BA$. Prove that $AB\in Y$.

(iv) Let $$Z:=\{A\in X: \operatorname{tr}(AB)\geq 0 \text{ for all } B\in Y\},$$ where tr means trace. It is a known fact that $\operatorname{tr}(AB)=\operatorname{tr}(BA)$ for matrices A,B with compatible sizes. Prove that $Z=Y$.

Problem 2.

(i) Let $X:=(-0.01,2\pi)\times(-0.01,2\pi)$. Consider $f:X\to R$ given by $f(x,y)=(\cos x)(\cos y)$. Find all its local maximals (both points and values) and local minimals (both points and values).

(ii) Let X be the set of real 2×2 orthogonal matrices. Consider $f:X\to R$ given by $A\mapsto \operatorname{tr} A^2$ where tr means trace. Find all its ~~go~~ global maximum and minimum (both points and values).

Problem 3. We define an equivalence relation $\sim$ on $M_n(k)$ by
$$A\sim B \text{ if } B=S^TAS \text{ for some invertible } S\in M_n(k).$$

(i) For $k=C$ and $n=2$, find a complete set of representatives for $M_n(k)/\sim$.

(ii) For $k=R$ and $n=2$, find a complete set of representatives for $M_n(k)/\sim$.

Problem 4. Let M be a finite inner product space over C. For an ordered basis B, let G_B be its Gram matrix. A pair of dual bases consists of an ordered basis $B=(b_1,\cdots,b_m)$ of M and another ordered basis $C=(c_1,\cdots,c_m)$ of M such that
$$\langle b_i,c_j\rangle_M=\delta_{ij}.$$
where $\delta_{ij}:=1$ if $i=j$ and $\delta_{ij}:=0$ if $i\neq j$.

(i) Let $b_1,\cdots,b_m$ and $c_1,\cdots,c_m$ be dual bases. Give a simple formula for an element $v\in M$ as a linear combination of $b_1,\cdots,b_m$.

(ii) Let $b_1,\cdots,b_m$ be an ordered basis of M. Prove that there exist $c_1,\cdots,c_m\in M$ such that $b_1,\cdots,b_m$ and $c_1,\cdots,c_m$ are dual bases.

(iii) Prove that, if B and C are dual bases, then G_B and G_C are inverses to each other.

(iv) Disprove that, if G_B and G_C are inverses to each other, then B and C are dual bases.

Problem 5. Consider square matrices over $\mathbb{C}$. It is a known fact that every matrix is upper triangularizable. Prove the following statements.

(i) Every matrix is unitarily triangularizable.

(ii) Eigenspaces with distinct eigenvalues of a normal matrix are orthogonal.

(iii) Every normal matrix is unitarily diagonalizable.

(iv) A matrix is unitarily diagonalizable if and only if it is normal.

Problem 6. Let $k := \mathbb{C}$. For a $A \in M_n(k)$, let $e^A := \sum_{i=0}^{\infty} \frac{A^i}{i!}$,

where we adopt the convention that $A^0 = I$. It is known facts that e^A always exists (the series converges) and $e^{A+B} = e^A + e^B$ for $[A,B]=0$. A logarithm of A is a matrix B such that $e^B = A$.

(i) Find all logarithms of the 2×2 identity matrix.

(ii) Let $\theta \in R$ such that $\sin\theta \neq 0$. Let $A_\theta := \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

Find all logarithms of A_θ.

(iii) Prove that, for a unitary matrix A, there exists a skew-Hermitian ($B^\dagger = -B$) logarithm B.

(iv) Prove that a logarithm of A exist if and only if A is invertible.

Problem 7. Let M be a finite k-module over an algebraically closed field k. Let $g := \mathrm{End}(M)$.

For each $x \in g$, let $x = x_s + x_n$ be its (unique) Jordan-Chevalley decomposition in g. For each $x \in g$, let $\mathrm{ad}\, x : g \to g$ be the (a priori not necessarily linear) map given by $y \mapsto [x,y]$, where $[x,y] := xy - yx$.

(i) Prove that the map and x is an element in $\mathrm{End}(g)$, i.e. and x is a linear transformation

(ii) Prove that the map $\mathrm{ad}\, x_s$ is semisimple in $\mathrm{End}(g)$.

(iii) Prove that the map $\mathrm{ad}\, x_n$ is nilpotent in $\mathrm{End}(g)$.

(iv) Prove that $[\mathrm{ad}\, x_s, \mathrm{ad}\, x_n] = 0$ in $\mathrm{End}(g)$.

Problem 8. Let $k := \mathbb{C}$, $d \geq 1$, and $c_0, \cdots, c_{d-1} \in k$. Let $p := x^d + c_{d-1}x^{d-1} + \cdots + c_1 x + c_0$ and suppose that $p = (x-\lambda_1)^{d_1} \cdots (x-\lambda_t)^{d_t}$ for distinct $\lambda_1, \cdots, \lambda_t \in k$. The companion matrix of p is

$$C := \begin{bmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & 0 & \vdots \\ 0 & 0 & \cdots & 1 & -c_{d-1} \end{bmatrix}_{d\times d}$$

(i) For $d=2$, find an explicit similarity from C to its Jordan normal form.

(ii) Consider a recurrence $f_0 = 0$, $f_1 = 1$ and $f_{n+2} = af_{n+1} + bf_n$, where $a, b, f_i \in \mathbb{C}$. Given an explicit formula (without using matrices) for f_n in terms of a, b, n.

(iii) Prove that the only annihilating polynomial of C of degree at most d-1 is 0.

(iv) Calculate the Jordan normal ~~from~~ form of C.